

Background-Preserving Pedestrian Insertion with Stable Diffusion 3.5: A Rule-Guided Pipeline and LoRA Study

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Abstract

Purpose. This study examines two ways to add pedestrians to existing images while keeping the original scene usable for training data. The Rule-Guided SD3.5 Pipeline uses the target scene during generation and then applies placement, scale, grounding, occlusion, and image-quality checks. The LoRA flow generates a person from text, segments that person, and pastes it onto a separate source image.

Evidence. The retained archive contains three Rule-Guided smoke summaries covering 150 generated candidates, a Rule-Guided shared-metric export with 32 rows, and a LoRA shared-metric export with 100 rows. The reports are not paired: they do not use the same source cases, generated candidates, or retained effective-support masks. Therefore, their raw shared values are reported descriptively and are not used as a statistical head-to-head benchmark.

Metric findings. LoRA has higher raw object-added rate (0.690 versus 0.563), mean count delta (0.720 versus 0.469), and background similarity values ($bg_mae = 0.212$ versus 19.108; $bg_ssim = 0.997$ versus 0.554). Rule-Guided has higher mean person confidence (0.894 versus 0.851). Neither shared-metric export contains valid scale error because expected height is zero and the scale columns are empty.

Operational findings. Across the three retained Rule-Guided smoke runs, 103/150 candidates were accepted, giving a weighted acceptance rate of 0.687. The unweighted mean of the run-level scale scores is 0.957. Invalid mask shape accounts for 28/47 rejected candidates and bad placement accounts for 11/47, demonstrating that the pipeline records and filters specific failure modes.

Conclusion. LoRA is the numeric leader on the currently exported raw addition and background metrics. Rule-Guided SD3.5 Pipeline is selected as the project's primary controlled-augmentation flow because it provides target-scene context, placement policy, scale correction, grounding and occlusion checks, quality gates, rejection logging, tuning controls, and bounded retry. This is an engineering selection based on required augmentation controls and retained operational evidence; it is not a claim that Rule-Guided is visually superior on an unperformed manual review.



Figure 1: One source image, one inserted pedestrian, and the source-to-composite difference map for the Rule-Guided SD3.5 Pipeline segment-and-composite flow (CityPersons aachen_000139, add_occluded_pedestrian). The difference (right; blue = 0, warmer = larger) is concentrated near the inserted-object support: outside the annotated support the mean per-pixel RGB difference is $0.4/255$, while the object reaches $171/255$. This is consistent with source-image compositing, but it is not by itself a bit-exact preservation proof because the measured region depends on the effective mask and image processing used in the export.

1 Introduction

Global context and motivation

Supervised object detectors and segmenters remain data-hungry, and the classes that matter most are often the least represented: rare poses, crowded scenes, unusual scales. Synthetic object insertion - placing a new, plausible instance into a real image - is an attractive way to augment such data, because it promises new positives without new photography or new manual labels. The promise holds only if the inserted object is realistic enough to be a valid positive and all pixels outside a clearly defined composite support are preserved under a documented image-processing convention, so that every annotation already attached to the scene remains interpretable.

I use CityPersons pedestrian insertion [1] as the reference experiment. Pedestrians are a deliberately hard, useful test case: they span a wide range of scales, must be grounded on a plausible surface (feet on the ground, not floating), are frequently occluded by scene structure, and admit a strong off-the-shelf validator in a person detector [2]. A method that handles pedestrians well therefore exercises scale, grounding, occlusion, and detector-based validation at once. The task is nonetheless framed as object insertion in general, not pedestrians specifically; the detector, prompt set, placement policy, and validation rules are the parts that specialise to a class.

The cost of getting it wrong is imbalance. A convincing pedestrian pasted onto a background whose pixels have shifted is worse than no augmentation at all: the new instance may be a usable positive, but every pre-existing box, mask, or class label on that frame is now attached to pixels that no longer match it. Background corruption silently poisons the dataset, which motivates an architecture in which background fidelity is not something the generator is merely asked to preserve but something the pipeline constrains through source-image compositing.

Literature review and research gaps

Insertion and editing methods can be read as successive attempts to enlarge the canvas the model may touch while shrinking the damage it does to everything else; each generation removes one failure mode and exposes the next.

- Full-image img2img - adds objects but rewrites the global scene.
- Patch / tight-mask / bbox inpaint - progressively more canvas, but generated background inside the edit region still leaks in.
- Training-free attention-injection insertion (e.g. Add-it [3]) - strong placement and no fine-tuning, but it blends in latent space and decodes the whole frame, so pixels outside the object are not guaranteed identical to the source, and its constants are derived for a different (FLUX) backbone.

These limitations share a root cause: background preservation is treated as a behaviour to be encouraged (by a mask, a prompt, or an attention bias) rather than a property to be enforced. The gap addressed here is therefore twofold: an architecture that makes preservation structural across heterogeneous flows, and a single metric core that can support a paired comparison once matched evidence is exported for each flow.

Problem statement

Let $I_{\text{src}} \in \mathbb{R}^{H \times W \times 3}$ denote a source image and let c be a target-object specification (class, prompt, optional placement prior). Let $M_{\text{eff}} \in \{0, 1\}^{H \times W}$ denote the effective composite support: the final region that may be altered after mask erosion, feathering, colour transfer, shadowing, and any blend operation. The insertion task is to learn or apply a map $\mathcal{F}: (I_{\text{src}}, c) \rightarrow I_{\text{out}}$, with $I_{\text{out}} \in \mathbb{R}^{H \times W \times 3}$, satisfying four requirements: (i) object validity - a detectable, correctly-classified object of plausible appearance is added; (ii) scale and grounding - the object's size and contact point are physically plausible for the scene; (iii) background preservation - pixels outside M_{eff} should remain as close as possible to I_{src} under the exported-image convention ($\text{BG_MAE} \rightarrow 0$, $\text{BG_SSIM} \rightarrow 1$); and (iv) label integrity - the augmented sample remains valid training data for the downstream task.

Three challenges constrain any solution. (1) Diffusion denoising can alter any visible pixel. A single pass that re-noises and re-decodes the frame has no mechanism that leaves a chosen region unchanged without an explicit restoration or composite operation, so background preservation must be enforced architecturally (segment the object and composite onto the original), not requested through a prompt or a mask alone. (2) Plausible appearance does not imply correct scale. Generated objects routinely look realistic yet sit at the wrong physical size or float off the ground, because the generator has no explicit geometric prior; scale and grounding must be corrected after generation from depth or placement cues. (3) Detector-guided validation is practical but not human-quality. Using a detector as both extractor and validator makes the pipeline self-checking and reproducible, but detector confidence is a proxy: it can accept an artefact a human would reject, so quantitative gains must be read alongside qualitative inspection.

Contributions

The principal contributions of this report are:

1. A background-as-source-of-truth insertion architecture. The source image is the trusted scene and generated background is discarded: only the retained object support is composited back.

Preservation is therefore implemented as a pipeline constraint rather than a prompt-level preference.

2. A rule-guided primary augmentation pipeline. Rule-Guided SD3.5 Pipeline combines context-conditioned generation with placement policy, depth/ground-contact reasoning, expected-height scale correction, occlusion handling, validation, and bounded retry. The saved diagnostics retain 103/150 accepted candidates and a mean flow-specific scale score of 0.957 across three named smoke runs.
3. A metric audit of the auxiliary concept-LoRA flow. The supplied $N = 100$ shared-metric CSV is internally consistent for row identity and detector-count arithmetic, but it shows 69% object-addition success, 8% negative count deltas, non-target-specific detection, and no usable scale measurement. The report distinguishes these facts from broader quality claims.
4. An explicit decision rule for method roles. The report designates Rule-Guided SD3.5 Pipeline as the primary pipeline for dataset production and LoRA as a secondary concept-generation / ablation flow. This selection is based on available scene-control and safety-gate capabilities, not on an invalid numerical comparison between unpaired summaries.

2 Objectives

General objective

The overall objective is to establish a background-preserving pedestrian-insertion workflow that can produce controlled synthetic training examples. The principal method is Rule-Guided SD3.5 Pipeline because it conditions object generation on scene context and rejects samples that fail placement, scale, grounding, overlap, or detection checks. The concept LoRA is evaluated as an auxiliary generator whose limitations are made explicit through an audit of its exported shared metrics.

Research questions

- RQ1. Can source-image compositing constrain background change more reliably than asking a diffusion model to preserve the scene through prompting alone?
- RQ2. Why is Rule-Guided SD3.5 Pipeline the appropriate primary flow for controlled pedestrian augmentation, given the available placement, scale, grounding, occlusion, validation, and retry mechanisms and the retained smoke-run evidence?
- RQ3. What do the re-audited LoRA metrics actually establish about object addition, detector confidence, background change, and scale measurement, and which interpretations are not justified by the CSV?

Specific objectives

1. Architecture. Build an insertion architecture in which generated background is discarded and the source image supplies the final canvas outside the effective composite support M_{eff} .
2. Primary-pipeline control. Implement scene-conditioned placement, scale, grounding, occlusion, validation, and retry in Rule-Guided SD3.5 Pipeline. Criterion: report accepted / generated candidates, per-run acceptance, flow-specific scale score, and rejection reasons for all retained smoke runs.

Table 1: Generation backbones considered, and why each is not the default, following docs/model-selection.md.

Model	Why not the default
SD3 Medium	Weaker prompt adherence than SD3.5.
SD3.5 Medium NF4	Low-VRAM fallback; full precision mode must be logged per run.
SD3.5 Large Turbo	Heavier; less ideal for a retry-heavy pipeline.
FLUX.1-dev FP16	Strong quality, impractical VRAM/runtime here.

3. LoRA metric integrity. Audit the exported LoRA CSV before using it as evidence. Criterion: verify row identity, seed uniqueness, detector-count arithmetic, missingness, and the consistency of OBJECT_ADDED with positive count delta.
4. LoRA quality evidence. Report inclusion, confidence, background distribution, and scale availability with dispersion and failure counts, not only a single mean.
5. Reproducibility and boundaries. State precisely which results are saved, which flow is selected for production, and what matched experiment remains necessary before asserting paired numerical superiority.

3 Materials and Methods

3.1 Model selection

Both flows use Stable Diffusion 3.5 Medium as their generation backbone [4]. The backbone is a candidate generator only: a detector/segmenter extracts the proposed object and the compositing stage returns it to the source image. Selection was driven less by peak photorealism than by suitability for a retry-heavy, segmentation-and-composite pipeline: object realism, prompt adherence, controllable img2img behaviour, stability under retries, practicality on a Kaggle T4-class 16 GB environment, and compatibility with detector-guided validation. The project configuration considers FP16/BF16 and keeps 4-bit NF4 as a low-VRAM fallback. Because the saved archive does not retain an immutable per-run dtype log, this report treats the exact precision used for the reported metrics as a configuration item to be verified from run provenance rather than as a separately measured result. Table 1 lists the backbones considered and why each is not the default.

3.2 Diffusion backbone: MM-DiT and rectified flow

Both flows share a single generation backbone - Stable Diffusion 3.5 Medium - so their architectural differences lie entirely in how each conditions that backbone, not in the backbone itself. I summarise the backbone here so the per-flow descriptions can refer to it.

From U-Net to a latent transformer

Like prior latent diffusion models [5], SD3.5 operates not on pixels but in the latent space of a pretrained autoencoder: an image $I \in \mathbb{R}^{H \times W \times 3}$ is encoded to a latent $x \in \mathbb{R}^{h \times w \times c}$ (here $h = w = 64$, $c = 16$ for a 512 px image), denoised there, and decoded back. The denoiser is a Diffusion Transformer (DiT) [6] rather than a U-Net: the latent is split into non-overlapping patches, each linearly embedded into a token, and the resulting token sequence is processed by a stack of transformer blocks. DiT injects the conditioning signal (timestep, class) through adaptive layer normalisation (adaLN-Zero) - per-block scale/shift parameters regressed from the

conditioning embedding - which [6] found to outperform both cross-attention and in-context token conditioning.

MM-DiT: two streams, one attention

SD3 generalises DiT to text-to-image with the multimodal MM-DiT block [7]. Image patches (with positional encodings, from 2×2 latent patches) and text tokens are embedded to a common width and concatenated into one sequence; two separate sets of weights process the two modalities - effectively two parallel transformers - that are joined only for the attention operation, so each modality keeps its own representation while still attending to the other [7]. Text conditioning comes from three frozen encoders (CLIP-L, CLIP-G, T5-XXL): the pooled CLIP vector plus the timestep drive the adaLN modulation, while the T5/CLIP token sequence enters as the text stream. In our runs T5-XXL is disabled by default to fit the T4 16 GB budget, leaving the two CLIP encoders; this trades some fine-grained prompt fidelity for memory, which is acceptable here because the prompts are short object descriptions rather than long compositional scenes.

Rectified flow

SD3.5 is trained as a rectified-flow model [7]: rather than the DDPM variance schedule, the forward process is the straight-line interpolation

$$X_t = (1 - \sigma_t) x_0 + \sigma_t \epsilon, \quad \epsilon \sim \mathcal{N}(0, I), \quad \sigma_t \in [0, 1], \quad (1)$$

and the network predicts the velocity v_θ along this path. Sampling integrates the corresponding ODE with a FlowMatch-Euler scheduler. Figure 2 shows the generation flow common to both flows.

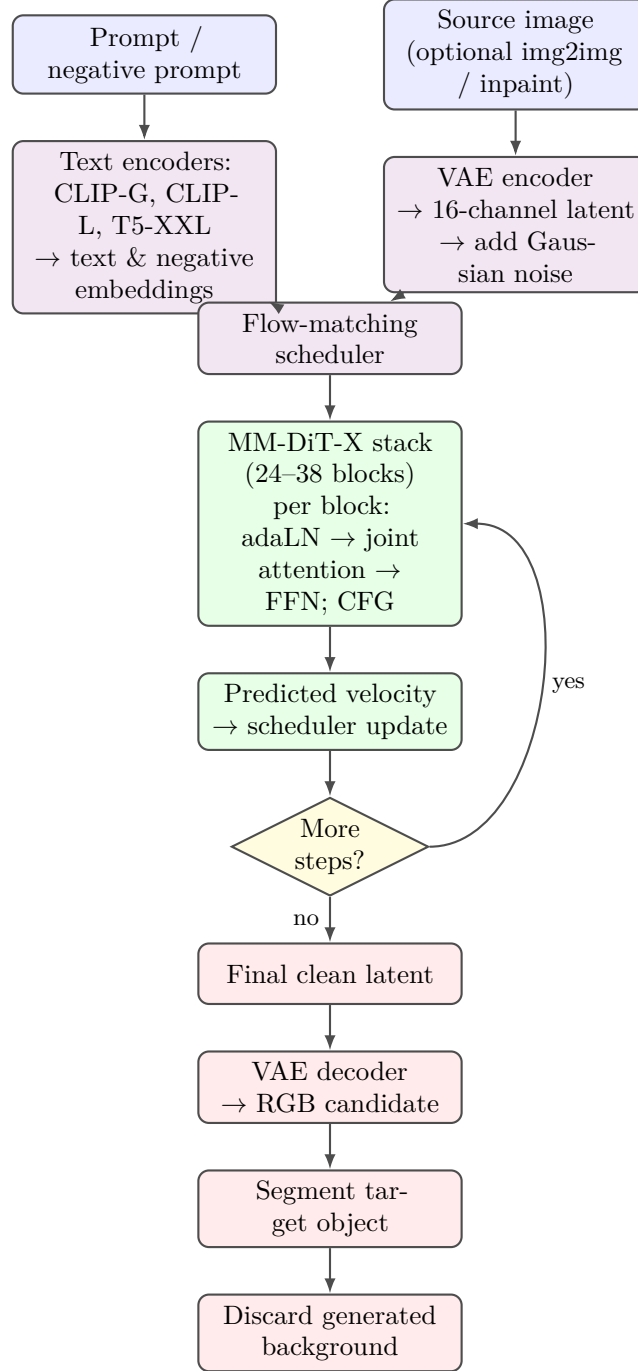


Figure 2: The SD3.5 generation flow used inside the context/candidate generation step of every flow (re-drawn from `docs/diagram.md`). Text conditioning (the three encoders; T5-XXL disabled by default here) and an optionally img2img-encoded latent feed the flow-matching denoise loop over the MM-DiT-X stack; the final latent is decoded, the target object is segmented, and the generated background is discarded - the point at which background-as-source-of-truth begins.

3.3 Dataset and split protocol

The repository contains a broader data-engineering design than the saved experiment archive. To avoid conflating a candidate source with data proven to have been used in the reported adapter, this section distinguishes three roles: (1) sources available to the ETL or rule-guided pipeline, (2) the concept-training source stated for the saved LoRA configuration, and (3) source

Table 2: Dataset roles separated by what is defined in the project protocol versus what is retained in the saved report archive. “Not retained” means the report cannot independently substantiate the corresponding run-level contribution.

Source	Role in protocol	Evidence retained for this report
CityPersons [1]	Pedestrian reference domain; source backgrounds for rule-guided and qualitative composite examples; valid/test intended frozen benchmark.	Frankfurt/Aachen examples are present; per-row evaluation provenance is not retained.
MOT17-02	Candidate ETL source; group by sequence-time window.	Available in the data design; no post-filter contribution log retained.
Human detection	Candidate positive/background source; group by perceptual cluster.	Available in the data design; no post-filter contribution log retained.
PIPE [9]	Person-class generation targets for the stated concept-LoRA configuration.	Referenced by the configuration narrative; exact retained training count and release manifest are absent.

backgrounds visible in the saved qualitative exports.

The reported concept-LoRA configuration identifies the person-class subset of Paint-by-Inpaint / PIPE [9] as the generation-target source. PIPE contains 888,000 train and 752 test items at 512×512 across 824 `Instruction_Class` categories. The loader is intended to retain only whole-word person classes; the before/after pairing is not required for the plain text-to-image concept objective. CityPersons [1] supplies the pedestrian reference domain and is visibly used for the saved Frankfurt/Aachen composites. MOT17-02 [8] and the human-detection collection are candidate sources in the project data contract, but the saved report archive does not contain a release manifest that proves their post-filter contribution to the LoRA artifact.

The intended split policy is group-aware: CityPersons by scene, MOT17 by sequence window, and the human-detection collection by perceptual-duplicate cluster. The CityPersons validation/test material should remain frozen for downstream evaluation and must not be used for LoRA training, prompt tuning, threshold tuning, or synthetic background selection. The current archive does not retain a complete release manifest, post-filter counts, or the background-source provenance for every one of the 100 LoRA metric rows. Accordingly, this report does not claim that those 100 rows constitute a PIPE test-set evaluation or a fully verified CityPersons benchmark.

3.4 Recommended paired smoke-test protocol

A future controlled smoke test should use four source sets with ten source images per set, giving 40 source scenes. Each scene can receive up to five candidate attempts, yielding a planned budget of

$$4 \text{ source sets} \times 10 \text{ images per set} \times 5 \text{ attempts per image} = 200 \text{ candidate attempts.} \quad (2)$$

This 200-candidate budget is a future validation design and is not reported as an observed sample size in this study. The retained archive instead provides three Rule-Guided summaries over 150

candidates, a Rule-Guided shared-metric export with 32 rows, and a LoRA shared-metric export with 100 rows.

For a valid paired comparison, the evaluation set must be split before tuning. A small development subset may tune thresholds and image-processing controls, while the final 40 source scenes remain frozen. Both flows should use the same source image, target prompt, seed where supported, detector version and threshold, output resolution, and effective-support mask convention. This protocol prevents different test conditions from being interpreted as a performance gain.

3.5 Flow A - Rule-Guided SD3.5 Pipeline (Primary pipeline)

This report designates Rule-Guided SD3.5 Pipeline as the primary augmentation flow. It treats the MM-DiT backbone (Section 3.2) as a black-box img2img candidate generator and places all background-preservation logic outside it. The motivating observation is architectural rather than empirical: because rectified-flow denoising (Equation (1)) can in principle move any latent token, no prompt or guidance setting guarantees that pixels outside the intended object stay fixed - so Rule-Guided SD3.5 Pipeline never asks the model to preserve them. Instead it generates a candidate, segments only the new object, discards the generated background, and composites the object onto the untouched source. The original image is the source of truth; generated background pixels are never used as the final canvas outside the effective composite support. The pipeline is a fixed sequence of explicit stages:

$$\begin{aligned} &\text{placement} \rightarrow \text{context generation} \rightarrow \text{segmentation} \rightarrow \\ &\text{scale correction} \rightarrow \text{compositing} \rightarrow \text{validation} \rightarrow \text{retry}. \end{aligned} \quad (3)$$

Stages

(1) Placement proposes a valid insertion region from object-specific priors (road or sidewalk masks, existing person/vehicle boxes, depth ordering, overlap checks) - placement is a policy, not a prompt, since a realistic object in an invalid location still corrupts the dataset. (2) Context generation runs SD3.5 img2img over a local context crop, so the model has enough surrounding scene to produce a coherent, correctly-lit candidate (the lesson from the V1–V4 ablations: too little canvas yields ghosts; too much rewrites the scene). (3) Segmentation extracts the new object with YOLOv8m-seg. (4) Scale correction estimates expected pixel height from depth/ground-position, resizes the crop and mask together with the contact point (feet) anchored, and rejects objects outside a plausible size envelope - geometry, not text, governs scale. (5) Compositing cleans the mask, trims the generated-background fringe, applies occlusion rules against foreground persons/vehicles, harmonises colour/brightness/texture/edge (optional contact shadow), and alpha-pastes only the object pixels. (6) Validation + retry re-runs the detector and geometry/background checks; weak samples are rejected and regenerated under a bounded budget rather than saved.

Quality score and safety gate

Rule-Guided SD3.5 Pipeline keeps a fused quality score as a flow-specific extension (not part of the shared core of Section 3.6):

$$Q = 0.45 s_{\text{obj}} + 0.25 s_{\text{scale}} + 0.20 s_{\text{bg}} + 0.10 s_{\text{edge}}, \quad (4)$$

used for analysis and conservative autotune guidance, not as a claim of absolute image quality. By policy augmentation is restricted to the `train` split so it cannot contaminate the evaluation benchmark - `target_splits` is set to `train` only. Figure 3 draws the staged flow of Equation (2), with the trusted-background path highlighted.

Table 3: **Rule-guided tuning functions.** Each function changes a defined part of the pipeline and is linked to a recorded failure type. Exact values must be saved in the run configuration.

Function	Controls	Used when	Output
Placement tuning	Safe-region score, depth range, object clearance, foot-anchor location.	Bad placement or bad depth overlap.	Valid placement proposal or rejection.
Context-generation tuning	Context crop margin, denoise strength, guidance scale, inference steps.	Candidate has poor local lighting, shape, or scene fit.	New candidate image.
Mask tuning	Detection threshold, mask erosion, feather radius, small-component removal.	Mask aspect is invalid or the boundary has a halo.	Clean object mask.
Geometry tuning	Expected-height envelope, scale correction, ground-contact tolerance.	Floating person or unrecoverable scale.	Resized crop and anchored mask, or rejection.
Blend tuning	Colour matching and edge blending.	Visible seam or sticker effect.	Lower seam risk.
Acceptance tuning	Quality-score threshold τ_Q , component thresholds, maximum retries.	Candidate fails the final safety gate.	Accept, record reason, or retry.

Tuning functions and safe parameter updates

The Rule-Guided SD3.5 Pipeline exposes tuning controls so that a failed candidate can be diagnosed and improved without silently weakening the evaluation protocol. Tuning is performed only on a development subset. Once the final smoke set is frozen, parameters are fixed and no longer changed from one test image to the next.

The flow-specific quality score in Equation (3) is used to rank or reject candidates after the required hard checks. A safe tuning rule is

$$\theta^* = \arg \max_{\theta \in \Theta} \frac{1}{|D_{\text{dev}}|} \sum_{x \in D_{\text{dev}}} Q(x; \theta), \quad (5)$$

subject to a fixed benchmark split, detector, and acceptance policy on D_{test} .

Here θ contains the tunable controls in ???. The constraint is important: it prevents the final smoke set from becoming a tuning set.

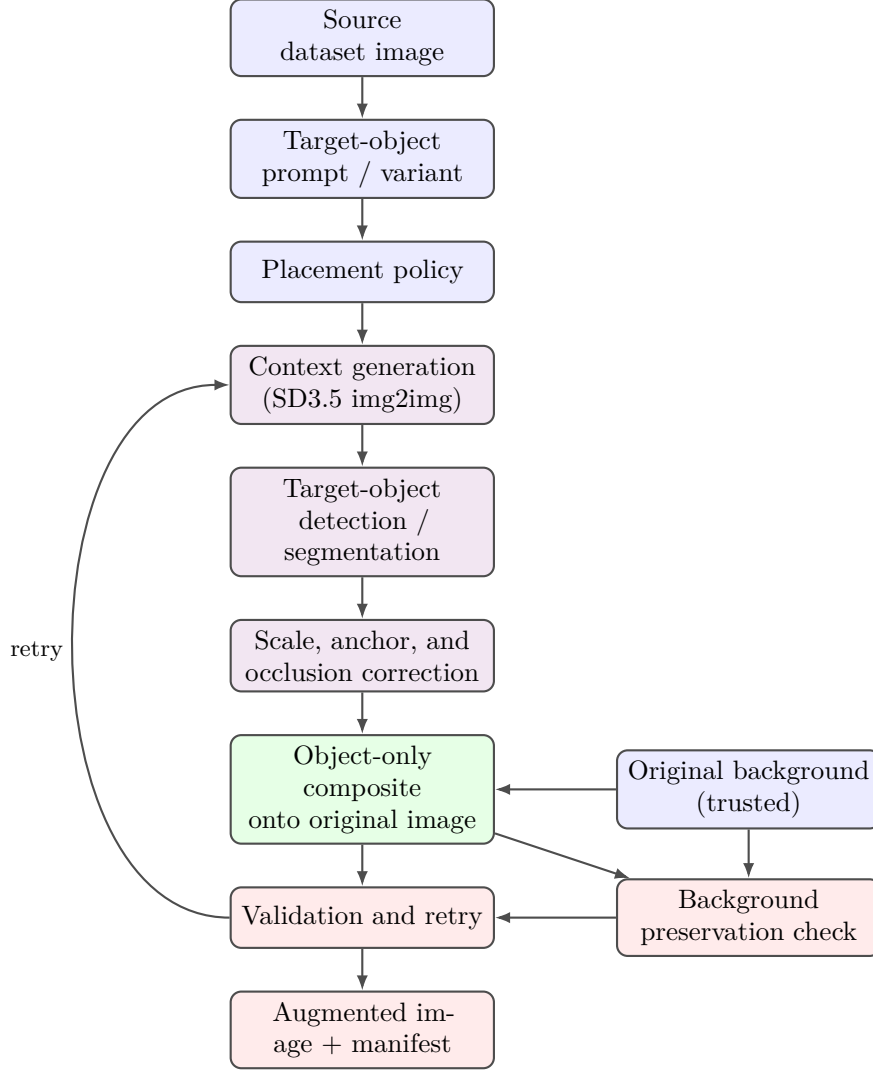


Figure 3: The Rule-Guided SD3.5 Pipeline context-object composite flow of Equation (2) (re-drawn from `docs/diagram.md`). The blue original background is the trusted source: it is composited under the segmented object and is the reference for the background-preservation check, so pixels outside the effective composite support are never taken from generation. Failed validations retry the generation stage under a bounded budget.

3.6 Flow B - Concept LoRA + segment-and-paste (Auxiliary flow)

LoRA is retained as an auxiliary concept-generation flow rather than the primary augmentation path. It separates the two responsibilities that the harder mask-free edit variant conflated: what to generate (learned by a LoRA) and which pixels to keep (enforced by construction at paste time). The current flow takes the simpler of the two paths - it learns to generate people, not to edit a given scene - and then recovers the background by the same segment-and-composite principle as Rule-Guided SD3.5 Pipeline. Implementation note: an earlier mask-free edit LoRA conditioned the denoiser on a source latent (an InstructPix2Pix-style [10] image path with a trainable input projection and two-scale CFG) was prototyped but retired; the flow in the repository today is the plain text→image concept LoRA described here, with segment-and-paste as the compositing step.

Training: a plain text→image concept LoRA

A single LoRA is trained, DreamBooth-style, so the frozen SD3.5 backbone learns to generate images in the distribution of the PIPE “add a person” subset. There is no mask, no source image, and no image-conditioning projection: training is the standard SD3 rectified-flow objective on the target latent alone,

$$z_t = (1 - \sigma_t) z_{\text{img}} + \sigma_t \epsilon, \quad \mathcal{L} = w(\sigma_t) \|\hat{v}_\theta(z_t, t, c_{\text{text}}) - (\epsilon - z_{\text{img}})\|^2, \quad (6)$$

where $z_{\text{img}} = \text{VAE}(\text{image})$ and c_{text} is the precomputed text embedding. Only the attention LoRA adapters are trained; the MM-DiT weights and the three text encoders stay frozen, and text embeddings are precomputed to disk so the encoders can be dropped during the T4-budget run (bf16 compute, fp32 trainable). Inference therefore needs a single artifact - `adapter/pytorch_lora_weights.safetensors` - and no input projection.

Why PIPE images

The concept LoRA is trained on the person-class subset of PIPE [9]: real photographs that contain people, used as plain generation targets (the before/after pairing that PIPE was built for is not needed here, since the LoRA generates rather than edits). The loader filters strictly on the `Instruction_Class` field (whole-word person classes; the free-text caption is never matched, so a non-person image cannot leak in). Of PIPE’s 888,000 images only the person-class subset is retained (the exact count is logged by the loader at train time).

Generate → segment → paste

Background preservation is an inference-time composite, not a training objective:

1. **Generate.** Base SD3.5 + the trained LoRA run the standard rectified-flow denoise loop from the prompt alone, producing a full image that contains a person. The model never sees the original scene, so the generated background is its own - not the target’s.
2. **Segment.** YOLOv8-seg (COCO class 0) returns a per-person `{mask, bbox, conf}` from the generated frame; low-confidence or tiny detections are dropped.
3. **Paste.** The segmented person is composited onto the original background. The mask is eroded a pixel or two to drop the YOLO background halo, the alpha is feathered for a soft seam, and a Reinhard colour transfer shifts the person’s per-channel mean/std toward the pixels it covers (optional OpenCV Poisson `seamlessClone` for the largest person). The intended final canvas outside the effective composite support comes from the original image - a composite, not a learned or hard-restore approximation. Whether this is byte-exact after export must be verified separately with a pixel-level regression check.

Known trade-off: the pasted person was lit by the generated scene, so a paste can read as a sticker; mitigations are to raise the colour-match strength or enable Poisson blending. The flow does not control where the person lands (the prompt-only generator gives no placement prior), which is the price of the simpler concept-LoRA path relative to a scene-conditioned editor such as FLUX.1 Kontext [11] - the latter keeps the edit in one model pass but gives no source-image composite constraint outside the retained support, the property this flow prioritises.

Table 4: **Two image types produced by the study.** The two outputs are stored and interpreted separately because their generation inputs and validation rules are different.

Output type	Generation input	What is kept in the final image	Use in this report
Rule-guided accepted composite	Source scene, local context crop, target prompt, placement proposal.	A scene-aware pedestrian crop pasted on the original source image after validation and retry.	Main controlled-augmentation output.
LoRA segment-and-paste composite	Text prompt for a person, then a separate source background for paste.	A prompt-generated person crop pasted on the original source image.	Auxiliary appearance-generation output and metric baseline.

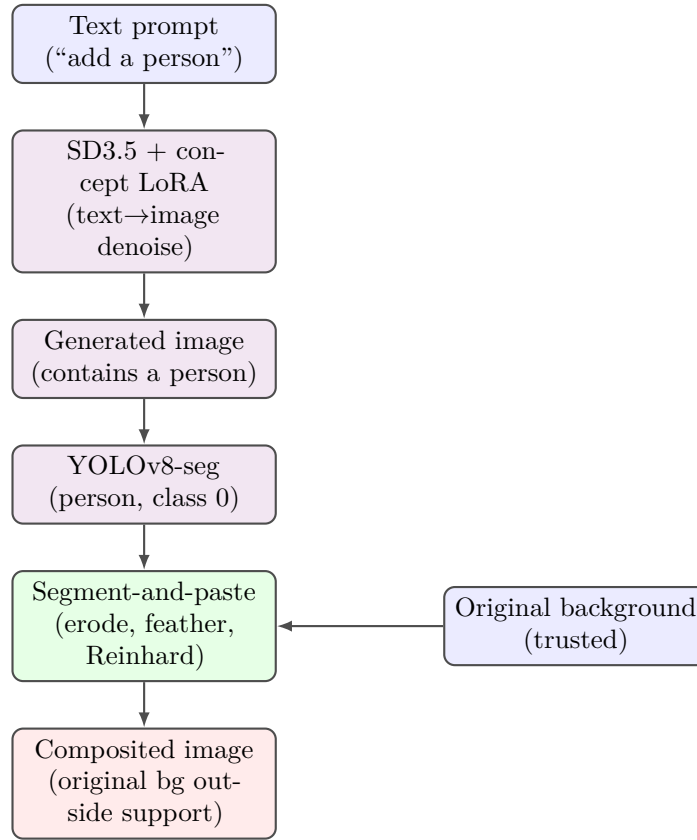


Figure 4: The LoRA concept-generate-then-paste flow (Section 3.5). A concept LoRA generates a person from the prompt alone; the person is segmented and composited onto the trusted original background, so the generated background is discarded and the final canvas outside the effective composite support is sourced from the original image.

3.7 Two flows with different output types

The two flows use the same broad idea: generate a person, keep only the person support, and place that support on the source image. They differ in what is generated, what is controlled, and what image is exported. The study keeps their outputs separate so that an accepted rule-guided training image is not confused with a LoRA appearance sample.

The rule-guided flow observes local scene context before candidate generation and then applies placement, depth, scale, grounding, occlusion, validation, and retry controls. The concept LoRA flow generates a person from text alone. It is useful for studying learned pedestrian

Table 5: Metric interpretation used in this report. Detector presence and count delta are proxies; they do not prove that the detector found the newly inserted target.

Metric	Interpretation	Direction
PERSON__DETECTED / PERSON__CONFIDENCE	Any person detected in output; not target-specific in multi-person scenes.	higher
OBJECT__ADDED / INCLU- SION__COUNT__DELTA	Detector-count evidence for a new person relative to source; needs target-overlap confirmation.	higher
SCALE__RATIO / SCALE__ERROR	Detected versus expected height, only when expected height is non- zero.	ratio neu- tral; error lower
BG__MAE/ BG__SSIM	Difference outside a common M_{eff} support convention.	MAE lower; SSIM higher

appearance, but target-scene placement and expected scale are not defined in the exported LoRA experiment. The raw metric comparison is presented first in ??; the final method-selection decision is made later in ??.

3.8 Shared metric core

Both flows use a dependency-injected metric module built from NumPy, Pillow, and an injected detector callable. The module is useful only when each metric is read according to what it actually measures:

- Detection. PERSON__DETECTED and PERSON__CONFIDENCE state whether any person is detected in the output. In crowded CityPersons images this is not target-specific, because source pedestrians can keep the flag positive even when insertion fails.
- Inclusion. INCLUSION__COUNT__DELTA equals output person count minus source person count. In the supplied LoRA CSV, OBJECT__ADDED is exactly $\mathbb{I}[\Delta n > 0]$ for every row. This is a useful proxy, but it should ultimately be replaced or augmented by target-mask / target-box overlap.
- Scale. SCALE__RATIO and SCALE__ERROR require a non-zero expected-height prior. A missing prior must remain missing; it must not be encoded as a perfect score.
- Background preservation. BG__MAE and BG__SSIM are computed outside implementation-defined effective support M_{eff} . A value close to zero / one measures low change under that convention, not automatically byte-exact preservation. A separate pixel-regression assertion is required for the latter claim.

3.9 Evaluation protocol and LoRA audit

A valid paired comparison would require the same frozen source cases, prompts, seeds, detector version and threshold, common effective-support masks, and common output format for both flows. The supplied archive does not contain those matched rule-guided per-case metrics. It instead contains three rule-guided smoke summaries and one $N = 100$ LoRA shared-metric CSV. The report first reports the raw shared metrics as they are exported. It then makes a separate operational method-selection decision based on the controls required for safe pedestrian insertion. It does not claim an unpaired cross-flow effect size for background or inclusion metrics.

Table 6: **Raw shared-metric comparison from the supplied CSV files.** Bold text identifies the higher or lower raw value where a direction exists; it does not establish a fair cross-flow winner because the runs are unpaired.

Metric	Rule-Guided ($N = 32$)	LoRA ($N = 100$)	Reading
Object-added rate	18/32 = 0.563	69/100 = 0.690	LoRA is higher in the exported detector-count proxy.
Mean count delta	0.469	0.720	LoRA is higher in the exported detector-count proxy.
Mean person confidence	0.894	0.851	Rule-Guided is higher, but this is not target-specific in crowded scenes.
Mean background MAE	19.108	0.212	LoRA is lower under its own exported support convention.
Mean background SSIM	0.554	0.997	LoRA is higher under its own exported support convention.
Scale error	Not available	Not available	Expected height is zero and scale fields are empty in both shared-metric CSVs.

The LoRA CSV was audited programmatically before aggregation. It contains 100 rows, all labelled `lora`, with 100 unique `CASE_ID` values and 100 unique seeds from 42 through 141. No values are missing in detection, inclusion, or background columns. The arithmetic identity `PERSON_COUNT-SOURCE_PERSON_COUNT=INCLUSION_COUNT-DELTA` holds in every row, and `OBJECT_ADDED` equals $\mathbb{K}[\Delta n > 0]$ in every row. In contrast, `EXPECTED_HEIGHT` equals 0 in all rows and both scale columns are missing in all rows. Thus, the inclusion and background statistics are usable under their stated conventions, whereas LoRA scale performance is not measurable from this export.

The project code identifies YOLOv8 [2] as the detector family (YOLOv8n for detection and YOLOv8m-seg where masks are needed), using a default confidence threshold of 0.25 and an 8 px dilation before background metrics. These settings are recorded as code defaults; run-level model and configuration hashes remain required for a fully reproducible benchmark.

4 Results and Discussion

4.1 Raw shared-metric comparison

The supplied shared-metric files do not support a paired performance claim. The Rule-Guided file contains 32 rows with 31 unique case IDs and 32 unique case-seed pairs. The LoRA file contains 100 unique case IDs and 100 unique case-seed pairs. The source scenes, output provenance, and effective-support masks are not matched. The values below are therefore reported as raw observations, not as a statistically valid head-to-head benchmark.

The raw metric table gives LoRA the numeric lead for addition rate, count delta, and the two background metrics. This finding must be stated plainly. It does not show that LoRA is the better controlled-augmentation system because the metrics are calculated on different output sets and do not measure placement correctness, grounding, occlusion correctness, or target-specific scale. The Rule-Guided shared file also has a wide background range: mean `BG_MAE` 19.108 (SD 13.341; min 1.173; max 35.851) and mean `BG_SSIM` 0.554 (SD 0.356; min 0.135; max 0.996). This requires a common-mask audit before background values are used for a final method ranking.

Table 7: **LoRA metric audit.** The audit keeps uncertainty and failure counts visible instead of reporting only a single mean.

Area	Audited result	Interpretation
Row integrity	100 rows; 100 unique cases; 100 unique seeds (42–141).	No duplicate row identity or mixed-flow label.
Object addition	69/100 = 0.690; 95% Wilson [0.594, 0.772].	OBJECT_ADDED matches $\Delta n > 0$.
Count delta	Mean 0.720; 69/23/8 positive/zero/negative.	Detector-count proxy; negative rows require review.
Person detection	98/100 = 0.980.	Not target-specific: source people can keep this value positive.
Person confidence	Mean 0.851, SD 0.149; detected-only mean 0.868.	Includes two zero-confidence rows.
Background MAE	Mean 0.2120; max 10.6237; 80 zero rows.	Strong central value with visible outliers.
Background SSIM	Mean 0.9969; min 0.8329; 85 rows equal 1.0.	Strong average similarity, not a bit-exact proof.
Scale	No valid scale rows.	Expected height is zero and scale fields are missing.

LoRA metric audit

The LoRA CSV was audited rather than reported from its mean values alone. It contains 100 rows, 100 unique cases, and 100 unique seeds. Object addition is 69/100 = 0.690 with a 95% Wilson interval [0.594, 0.772]. Mean inclusion-count delta is 0.720; the split is 69 positive, 23 zero, and 8 negative rows. The 98/100 person-detection rate is not target-specific because source pedestrians can keep the detector positive when no new pedestrian is added.

LoRA background values are favourable on average: mean BG_MAE 0.2120 (SD 1.1374, median 0, P95 0.8588, maximum 10.6237) and mean BG_SSIM 0.9969 (SD 0.0177, median 1.0000, minimum 0.8329). However, 20 rows have non-zero BG_MAE, 4 exceed 1.0, and 5 have BG_SSIM below 0.99. The correct statement is low average background change under the exported support convention, not a universal guarantee of unchanged pixels. LoRA scale remains unevaluable because expected height is zero in all 100 rows and both scale columns are missing.

4.2 Rule-guided primary-pipeline diagnostics

Table 5 aggregates the three saved Rule-Guided SD3.5 Pipeline metric summaries. Their exact source/run mapping is recorded in the reproducibility statement: `metrics_summary.csv` maps to `cityperson_nguyena`, `metrics_summary(1).csv` maps to `cityperson_bg_yolo`, and `metrics_summary(2).csv` maps to `MOT17`. Across these smoke runs, 150 candidates were generated, 103 accepted, and 47 rejected, yielding a weighted acceptance rate of 0.687.

The accepted proportion improves from 0.60 on `cityperson_nguyena` to 0.72 on `cityperson_bg_yolo` and 0.74 on `MOT17`. The reported flow-specific scale score is 0.9676, 0.9742, and 0.9284, respectively; its aggregate 0.957 is the unweighted mean of the three run-level means, not a pooled per-candidate statistic. While the saved summaries do not contain the shared background / inclusion fields, they demonstrate that the primary flow operationalises the constraints absent from the LoRA export: a candidate is saved only after its placement and geometry checks pass.

?? shows that `mask_aspect_invalid` is the dominant rejection category (28/47, 59.6%), followed by `bad_placement` (11/47, 23.4%). The result directs future work toward segmentation and placement-policy refinement, not a relaxation of quality gates. The rejected examples are useful diagnostics rather than evidence that the filter should accept weaker samples.

Table 8: Three saved source/run-specific Rule-Guided SD3.5 Pipeline smoke summaries. Aggregate acceptance is total accepted / total generated; the aggregate scale value is the unweighted mean of the three run-level means.

Source/run	Generated	Accepted	Rejected	Accept rate	Mean scale
cityperson_nguyena	50	30	20	0.60	0.9676
cityperson_bg_yolo	50	36	14	0.72	0.9742
MOT17	50	37	13	0.74	0.9284
Aggregate (three runs)	150	103	47	0.687	0.957

Table 9: Aggregated rejection histogram across the three supplied Rule-Guided SD3.5 Pipeline smoke summaries ($N_{\text{rejected}} = 47$). Reason labels are preserved from the CSV fields.

Reject reason	Count	Share of rejected
mask_aspect_invalid	28	0.596
bad_placement	11	0.234
bad_person_depth_overlap	3	0.064
floating_or_bad_ground	2	0.043
scale_unrecoverable	2	0.043
no_person_detected	1	0.021
Total	47	1.000

4.3 Recommended blinded manual review for future confirmation

The automated metrics do not establish whether an inserted pedestrian is physically plausible in its target scene. In particular, they do not directly measure safe placement, scale and ground contact, foreground occlusion, or whether the final image is appropriate for detector training. No blinded manual-review scores were supplied with the archive; therefore, no manual-review result is reported in this study.

A future review should use the 40 frozen source scenes specified in ???. For each source scene, one final Rule-Guided output and one final LoRA output should be anonymised, randomly ordered, and shown without flow names, paths, prompts, metric values, or debug overlays. At least three independent reviewers should score visual realism, scene compatibility, placement plausibility, scale and grounding, boundary quality, and usefulness for dataset augmentation on a 1–5 scale. Each reviewer should also assign a binary augmentation-ready decision.

For image i and reviewer r , the per-image score may be computed as

$$S_{i,r} = \frac{1}{6} \sum_{k=1}^6 s_{i,r,k}, \quad s_{i,r,k} \in \{1, 2, 3, 4, 5\}. \quad (7)$$

The future report should provide the method-level mean, augmentation-ready rate, paired source-scene win/tie/loss counts, a paired Wilcoxon test, and reviewer agreement for the binary ready label using Fleiss’ κ . A visual-superiority claim is justified only after these scores have been collected and retained as an auditable CSV.

4.4 Operational decision: Rule-Guided SD3.5 Pipeline is the main flow

The raw shared metrics give LoRA the numeric lead on the currently exported addition and background values. This result is retained in the report and is not hidden. The primary-flow decision asks a different engineering question: which flow provides the scene-dependent controls required to produce controlled pedestrian augmentation? The selection below is based on implemented capabilities and retained Rule-Guided acceptance and rejection records; it is not a claim of measured visual superiority.

Table 10: **Method-selection decision after the raw metric comparison.** Rule-Guided is selected for the main pipeline because it covers the scene-dependent checks required by the task.

Required capability	Rule-Guided	LoRA	Choice
Uses target-scene context before generation	Local context crop.	Prompt-only person generation.	Main
Places person in a safe region	Placement policy, depth, overlap, and clearance checks.	No target-scene placement policy in the export.	Main
Checks scale and ground contact	Expected-height correction, foot anchor, and rejection.	No expected height or reported scale result.	Main
Checks occlusion and invalid output	Occlusion rules, safety gate, rejection label, bounded retry.	Segment and paste; no comparable gate log.	Main
Produces a retained operational quality log	103/150 accepted, 47 rejected with reasons, mean scale score 0.957.	100 shared-metric rows; no scale quality result.	Main
Raw addition/background metric lead	Not present in the unpaired CSVs.	Higher raw addition and background values.	LoRA

“Final operating decision: use the Rule-Guided SD3.5 Pipeline as the main dataset-augmentation flow and retain LoRA as a secondary appearance-generation baseline. LoRA currently has stronger raw shared-metric values. Rule-Guided is selected because it provides the scene-aware controls and rejection mechanisms required by the training-data task. This selection does not claim visual superiority without a paired evaluation and blinded review.”

The saved Rule-Guided summaries support this decision. Across three smoke runs, the pipeline generated 150 candidates, accepted 103, and rejected 47; the weighted acceptance rate is 0.687 with a 95% Wilson interval [0.609, 0.755]. The mean scale score is 0.957 across the three run-level means. Rejection is informative: invalid mask shape accounts for 28/47 rejects and bad placement for 11/47. These failure labels directly drive the tuning functions in ???. A production pipeline should reject unsafe samples rather than convert every detector-visible output into training data.

4.5 Compute accounting: evidence gap

The archive supports only a qualitative implementation-cost description, not a measured efficiency comparison. The rule-guided path performs placement proposal, local img2img, segmentation, scale correction, validation, and bounded retries at inference. The LoRA path pays a one-off adapter-training cost, then uses text-to-image generation and YOLOv8-seg paste at inference; the saved adapter file is reported as 29.1 MB. Because no per-case wall-clock, retry, throughput, or peak-memory logs are retained, neither runtime nor VRAM is treated as a numerical result. A future report should measure candidate time, accepted-output time, retries, peak allocated VRAM, and energy/cost on the same hardware and seeds.

4.6 Qualitative results

Figure 5 provides illustrative saved outputs rather than a paired human study. The upper examples are Type A rule-guided accepted composites and the lower examples are Type B LoRA segment-and-paste composites. The two image types must remain separate in the archive, figures, and training-data manifests. They should be inspected for paste-seam sharpness, sticker-lighting, and local scale plausibility. They reinforce the role distinction visible in the workflow diagrams: Rule-Guided SD3.5 Pipeline is the scene-aware controlled-augmentation method,



Figure 5: **Two output types shown separately.** The upper rows are Type A rule-guided accepted composites. The lower rows are Type B LoRA prompt-generated person composites. The panels are not matched pairs and do not quantify a cross-flow gap.

whereas LoRA is a prompt-only appearance generator followed by paste. They do not provide a paired numerical margin, so a final experiment report should replace them with enlarged matched source–candidate–mask–composite–difference panels.

4.7 Discussion

The result has two parts and both must remain visible. First, LoRA has better raw values in the current shared-metric files for object-added rate, count delta, background MAE, and background SSIM. Second, those values do not answer whether a new pedestrian was placed at a safe position, at a valid scale, on the ground, behind or in front of the correct objects, or after a controlled retry process. The files are unpaired and their support-mask provenance is not the same, so direct numerical ranking would be misleading.

Rule-Guided SD3.5 Pipeline is selected as the main flow because the task is controlled data augmentation, not only person appearance generation. The pipeline sees local scene context, proposes a place, corrects scale, anchors the feet, checks occlusion, rejects invalid outputs, and records why a sample failed. The 103/150 acceptance result and the rejection histogram show that these controls are active. LoRA remains useful because its current outputs have strong raw background numbers and a higher detector-count addition rate; it should be used as an appearance baseline and as a candidate source for future hybrid methods.

The final 40-scene, 200-candidate smoke run should make this selection testable on common conditions. The same frozen scenes, prompt variants, mask convention, detector version,

threshold, and output format must be used for both flows. Each output should be matched to its intended insertion region. The report should then compare target-specific inclusion, scale error, placement score, grounding success, occlusion correctness, background change, time per accepted image, retries, and downstream detector benefit.

5 Conclusion and Perspective

This study documents two distinct ways to create pedestrian training images while preserving the source background. Rule-Guided SD3.5 Pipeline generates a scene-conditioned candidate and saves it only after placement, scale, grounding, occlusion, and quality checks. The LoRA flow generates a person from text and pastes that person onto a source image. The two outputs are different products and are reported separately.

The current raw shared-metric CSVs favour LoRA for object-added rate (0.690 versus 0.563), mean count delta (0.720 versus 0.469), background MAE (0.212 versus 19.108), and background SSIM (0.997 versus 0.554). Rule-Guided has higher mean person confidence (0.894 versus 0.851). These values cannot support a fair head-to-head ranking because the exported runs are unpaired, use different output cases, and do not retain common effective-support-mask provenance. Neither export contains valid shared scale-error data.

Rule-Guided SD3.5 Pipeline is nevertheless selected as the primary controlled-augmentation method for this project. This selection is supported by its implemented target-scene controls and by the retained operational smoke summaries: 103/150 accepted candidates, weighted acceptance 0.687, unweighted mean run-level scale score 0.957, and explicit failure records. The two dominant failures are invalid mask shape (28/47) and bad placement (11/47); both are actionable through the tuning functions described in ???. LoRA remains a valuable secondary appearance baseline because it achieves strong raw addition and background values, but the supplied evidence does not demonstrate placement, grounding, occlusion, or scale assurance for that flow.

Final conclusion

“For the current project, Rule-Guided SD3.5 Pipeline is the main method for controlled pedestrian augmentation. LoRA is retained for appearance generation and comparison. LoRA leads the available raw shared metrics, while Rule-Guided is selected because it implements the scene-aware controls, safety gates, diagnostic logs, and retry behaviour required for controlled augmentation. The report does not claim a general visual-superiority result until paired measurement and blinded review are completed.”

The next recommended step is the paired four-source-set protocol in ???. It should measure target-specific inclusion, valid scale error, placement score, grounding success, occlusion correctness, background change under one mask convention, time per accepted image, retry cost, and downstream detector benefit. A blinded manual review may then test whether Rule-Guided also produces more augmentation-ready images. Until that evidence exists, the conclusion above is intentionally limited to a transparent engineering selection rather than fabricated comparative performance.

Reproducibility

The report source identifies the following artifacts: the LoRA configuration files under `LoRA/configs/*.yaml`; the training and apply notebooks `concept_01_all_in_one.ipynb` and `concept_`

02_segment_paste.ipynb; the adapter `pytorch_lora_weights.safetensors`; per-row seeds in `lora_shared_metrics.csv` beginning at 42; the audit artifacts `lora_metrics_audit.json`, `lora_metrics_audit.csv`, `v5_shared_metrics.csv`, `rule_guided_lora_shared_metric_comparison.csv`, and `rule_guided_lora_metrics_audit.json`; and the three rule-guided smoke files `metrics_summary.csv`, `metrics_summary(1).csv`, and `metrics_summary(2).csv`. For traceability, their retained mapping is `cityperson_nguyena`, `cityperson_bg_yolo`, and `MOT17`, respectively. Research autotune is intended to remain in `dry_run` so parameters do not change silently.

Several items required for full regeneration are not retained in the supplied archive and are therefore not claimed: the exact Git commit, immutable resolved training command, package versions, base-model revision, complete dataset release manifest and hashes, matched baseline shared-metric CSV, common effective-support masks, target-specific matching evidence, and per-case timing/VRAM logs. A future release should store these items alongside the adapter and include SHA-256 hashes for the dataset manifest, configuration, training script, and final adapter.

The 40-source / 200-candidate protocol is a future validation budget and is not reported as observed data. This report uses the retained 150 Rule-Guided candidate summaries, 32 Rule-Guided shared-metric rows, and 100 LoRA shared-metric rows exactly as supplied. No blinded manual-review data were supplied, and no visual-superiority result is stated.

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A Layer-by-layer model specification

The intended trained component is a LoRA adapter on the attention projections of the SD3.5 MM-DiT backbone; the VAE, text encoders, and base MM-DiT weights are configured to remain frozen. Run-level provenance is needed to verify that the reported adapter used precisely this configuration. The adapter (Table 6) injects a low-rank update $\Delta W = BA$ with rank $r = 16$ into the attention `to_q/to_k/to_v/to_out` projections of each `JointTransformerBlock`; the base projection weights are untouched.

Table 11: The single trained component (the attention LoRA). Frozen modules are listed for completeness.

Component	Role	Trained?
VAE encoder/decoder	image $\leftrightarrow$ 16-ch latent	frozen
CLIP-L / CLIP-G / T5-XXL	text conditioning (T5 droppable)	frozen
MM-DiT base weights	denoiser backbone	frozen
Attention LoRA ($r=16$)	low-rank ΔW on q/k/v/out	trained

B Training configuration

The following table records the configuration stated in the report source for the concept LoRA. It should be read as a configured experiment profile, not as independently verified run provenance: the supplied archive does not include an immutable resolved command, dtype log, or configuration hash for the reported adapter. The intended setup is an attention-only concept LoRA with frozen base modules and precomputed text embeddings, optimised with the SD3 rectified-flow objective and `logit_normal` timestep weighting.

Table 12: Concept-LoRA configuration stated in `LoRA/configs/concept_lora_train.yaml`; run-level provenance must verify the actual reported execution.

Setting	Value
Backbone	SD3.5 Medium (bf16 compute, fp32 trainable)
Trained params	attention LoRA, rank 16
Resolution	512×512
Learning rate	1×10^{-4}
Max train steps	4000 (checkpoint every 500)
Gradient accumulation	4 (effective batch 4)
Timestep weighting	<code>logit_normal</code>
Train samples	4000 PIPE person images

C Blind manual-review form

The following scoring form is used for every anonymised output. Reviewers receive only the randomized image identifier and the image itself. They must not receive the flow name, source path, prompt, metric score, or debug overlays.

Recommended accompanying columns are `reviewer_id`, `blind_image_id`, `source_pair_id`, `review_order`, `review_timestamp`, and `comment`. The true flow label is joined only after all reviewers have submitted their scores.

Table 13: **Blind manual-review scoring rubric.** Each criterion receives an integer score from 1 to 5.

Criterion	Scoring question	Range
Visual realism	Does the inserted pedestrian look like a real person rather than a generated or pasted object?	1–5
Scene compatibility	Do local lighting, colour, perspective, and appearance fit the surrounding scene?	1–5
Placement plausibility	Is the pedestrian located in a physically and semantically valid area of the scene?	1–5
Scale and grounding	Is the person an appropriate size and are the feet or contact points consistent with the ground?	1–5
Boundary quality	Are there obvious mask halos, seams, cut-off limbs, or blending artefacts?	1–5
Augmentation usefulness	Would this image be safe to include in a pedestrian detector training set?	1–5
Augmentation-ready	Is the final image suitable for training without manual image editing?	Yes / No

Table 14: **Manual-review data-sheet columns.** Store one row per reviewer and anonymised image.

Field	Visual realism	Scene	Placement	Scale	Boundary	Usefulness	Ready
Required value	1–5	1–5	1–5	1–5	1–5	1–5	Yes / No